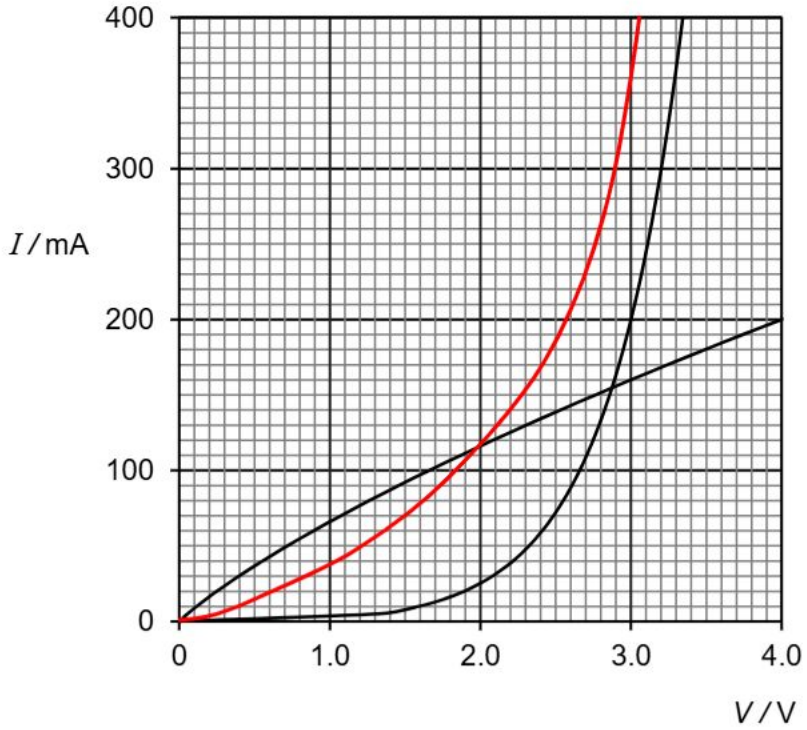
**6** **(a)** **(i)** $\text{e.m.f} = 3.00 + 4.00$

$= 7.00$

**(ii)** $R = \dfrac{3.00}{0.200} = 15.0\ \Omega$

**(b)** **(i)**




*smooth curve starting from the origin with increasing gradient

**(ii)** As the current (or p.d.) increases, joule heating causes more charge carriers to be released in the thermistor but not in the resistor. Hence, the resistance of the thermistor decreases with current (or p.d.).